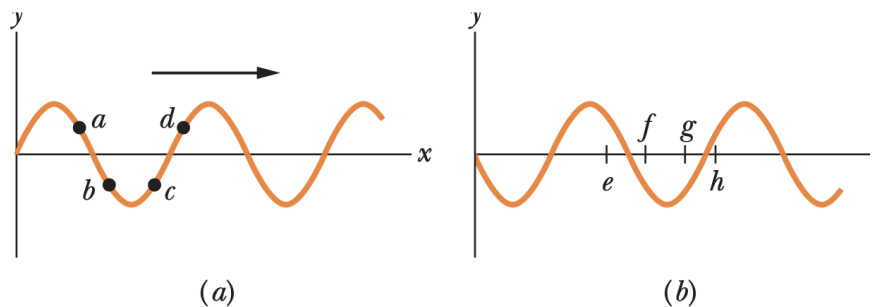


The Wave Equation and Traveling Solutions

1 *Up-And-Down, There Goes the Wave [Quick]

The figure (a) below gives a snapshot of a wave traveling in the direction of positive x along a string under tension. Four string elements are indicated by the lettered points.

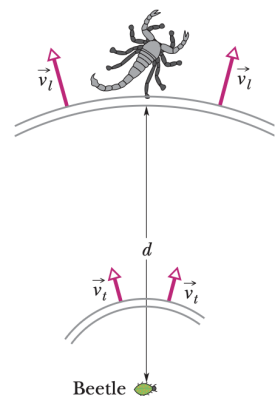
- For each of those elements, determine whether, at the instant of the snapshot, the element is moving upward or downward or is momentarily at rest. *Hint: Imagine the wave as it moves through the four string elements, as if you were watching a video of the wave as it traveled rightward.*
- Figure (b) gives the displacement of a string element located at, say, $x = 0$ as a function of time. At the lettered times, is the element moving upward or downward or is it momentarily at rest?



2 *The Beetle and The Scorpion [Quick]

A sand scorpion can detect the motion of a nearby beetle (its prey) by the waves the motion sends along the sand surface. The waves are of two types: transverse waves traveling at a speed of v_t and longitudinal waves traveling at v_l with $v_l > v_t$.

- If a sudden motion sends out such waves, a scorpion can tell the distance of the beetle from the difference Δt in the arrival times of the waves at its leg nearest the beetle. In terms of v_t , v_l and Δt , compute an expression for the distance d .
- The Indian Thar Desert is home to the *Hottentotta tamulus* scorpion (aka the Indian Red Scorpion), one of the most medically dangerous scorpions in the world. The *H. tamulus* likes to feed on the Darkling Beetle also abundant in the Thar desert. If, for the Thar soil, $v_t = 50$ m/s and $v_l = 150$ m/s, and the scorpion registers a 4 millisecond delay, how far would it have to jump on to get the beetle?

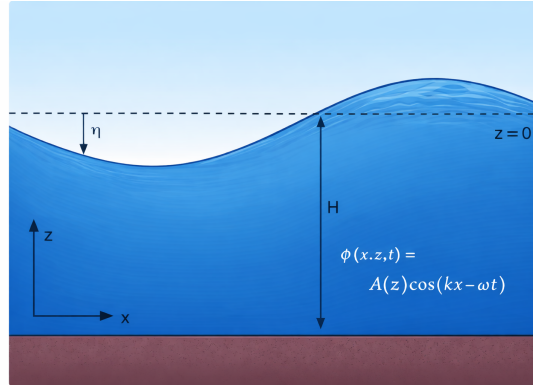


3 *Ocean Waves in Deep Water [Substantive]

In class we discussed sinusoidal plane wave solutions of the form $y(x, t) = Ae^{i(kx - \omega t)}$ where $\omega = 2\pi f$ is the angular frequency and $k = 2\pi/\lambda$ is the wavenumber. The **dispersion relation** $w \equiv w(k)$ connects the two. For plane waves, we found $\omega = vk$ where v is the wave speed, which turned out to be a constant for each wavelength in the case of plane waves. Let us now apply the same kind of ideas to the somewhat more complex, real-world scenario of water waves propagating on the surface of a deep ocean.

Ocean waves are one of nature's most familiar yet subtle phenomena. A single wave crest traveling across

the open sea carries within it a rich interplay of gravity, fluid inertia, and geometry. See this simulation of ocean waves: <https://www.acs.psu.edu/drussell/Demos/waves/wavemotion.html> (it's towards the second half of the page). These waves are a combination of both longitudinal and transverse motion. Further, not all waves travel at the same speed — the speed of a wave depends on its wavelength, a property known as dispersion. This has striking consequences: a distant storm generates waves of many wavelengths simultaneously, but by the time they reach a shoreline hundreds of kilometers away, the long swells have outrun the short ripples, arriving in an orderly sequence sorted by wavelength. We will discuss this in detail later in the class, but *today I want you to focus only on getting the dispersion relation for such ocean waves.*



We consider waves traveling along the x direction on the surface of water. The vertical coordinate z measures height, with $z = 0$ representing the undisturbed surface of the water. The water occupies the region $z < 0$. The vertical displacement of the surface is denoted $\eta(x, t)$. A convenient way to describe the motion of the water is through a scalar function $\phi(x, z, t)$, called the **velocity potential**, from which the velocity of the fluid is obtained by taking its partial derivatives (we won't prove this):

$$\vec{v} = \underbrace{\left(\frac{\partial \phi}{\partial x}\right)}_{v_x} \hat{i} + \underbrace{\left(\frac{\partial \phi}{\partial z}\right)}_{v_z} \hat{k}.$$

Since water waves consist of each element having both transverse and longitudinal components, there is velocity both in the x and z directions. For an incompressible fluid, the velocity potential satisfies the following equation (which will be our equivalent of the wave equation here),

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial z^2} = 0. \quad (1)$$

Observations show that waves travel along the surface with a definite **wavenumber** k and **angular frequency** ω . We therefore try a wave-like form for the velocity potential,

$$\phi(x, z, t) = A(z) \cos(kx - \omega t)$$

where the function $A(z)$ describes how the motion varies with depth. As you would expect (or see from the simulation) that waves deeper in the water don't move as much, so amplitude A should be a function of depth z .

- Show that the horizontal velocity component is $v_x = -kA(z) \sin(kx - \omega t)$. Also determine the vertical velocity component v_z . You can leave it in terms of $dA(z)/dz$ if needed.
- Substitute the ansatz $\phi(x, z, t) = A(z) \cos(kx - \omega t)$ into our “wave equation” of Eq. (1) to show that $A(z)$ satisfies

$$\frac{d^2 A(z)}{dz^2} - k^2 A(z) = 0.$$

- Solve this differential equation to obtain the general form of $A(z)$. Think exponentials.
- We now consider **deep water**, meaning the ocean depth H is much larger than the wavelength, *i.e.* $H \rightarrow \infty$. Physically, the motion of the fluid should become negligible far below the surface. Use this condition to determine which of the mathematical solutions for $A(z)$ is physically acceptable.

The vertical motion of the surface must match the vertical velocity of the fluid at the surface:

$$\left. \frac{\partial \eta}{\partial t} = \frac{\partial \phi}{\partial z} \right|_{z=0}.$$

- e Assume the surface displacement has the form $\eta(x, t) = B \sin(kx - \omega t)$. Now use your expression for v_z from part (a) enforce this surface matching condition above to determine the relation between B and A_0 , where $A_0 = A(0)$.

The pressure at the free surface must equal atmospheric pressure. For small waves this leads to the linear Bernoulli condition (again, we won't prove it here),

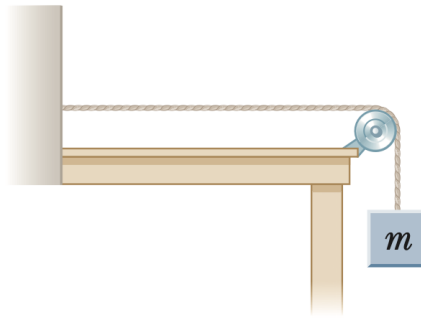
$$\frac{\partial \phi}{\partial t} + g\eta = 0 \quad \text{at } z = 0.$$

You can think of this in the following way: the time derivative of the velocity potential measures the acceleration of the fluid. At the surface, that acceleration must balance the gravitational pull on the displaced surface element.

- f Use this Bernoulli condition to show that the ocean wave must satisfy the dispersion relation $\omega = \sqrt{gk}$.
- g Remembering that the speed of a particular wave is given by ω/k show that an ocean wave of wavelength λ has a speed $\sqrt{g\lambda/(2\pi)}$. What does this imply about how the speed of ocean waves depends on their wavelength?

4 More Waves on Strings, on Earth and on the Moon [Simple]

Consider the following setup, where a string passes over a pulley and support a mass m . If the mass of the string is m_{st} and has a length l , find the following quantities:



- a In terms of the given variables m , m_{st} , l and acceleration due to gravity g on Earth, find an expression for the speed of a wave on this string.
- b If I excite a sinusoidal wave on this string of frequency f , what will be the corresponding wavelength of the wave in terms of f and other variables you used in part (a)?
- c If the mass hanging is $m = 2$ kg, and the string has a mass of $m_{\text{st}} = 4$ g and total length of 1.6 m, how long will it take for a wave pulse to cover the horizontal section of the string which has a length of 1.3 m?
- d How would your answers change if you were doing this experiment on the moon where the gravitational force is roughly 1/6th that of the Earth?
- e And finally, a conceptual question. What if the block were swinging back and forth with respect to the vertical like a pendulum? How would that affect the wave speed on the string?

5 Waves on a Rope [Experiment]

In this simple experiment you will study how waves propagate along a stretched rope and use your measurements to estimate the tension in the rope. Take a rope from the TA. Two students hold the ends of a nylon rope of length $L = 1.5$ m so that it is stretched horizontally. The rope has total mass of about $M = 6$ g. One student generates small transverse pulses by flicking the rope up and down. Use the *slow-motion* video mode on your phone to record the motion of the pulse as it travels along the rope.

- a From the video, estimate the time t taken for a pulse to travel from one end of the rope to the other. Estimate the wave speed using $v = L/t$.
- b Repeat the experiment for two cases. Measure the travel time and estimate the wave speed in each case:
- Case 1: The rope held moderately tight.
 - Case 2: The rope pulled as tight as possible.

Case	Travel time t	Wave speed v
Lower tension		
Higher tension		

- c Do your results agree with the prediction that wave speed increases with tension?
- d Use your measured wave speeds to estimate the tension in the rope for each case.

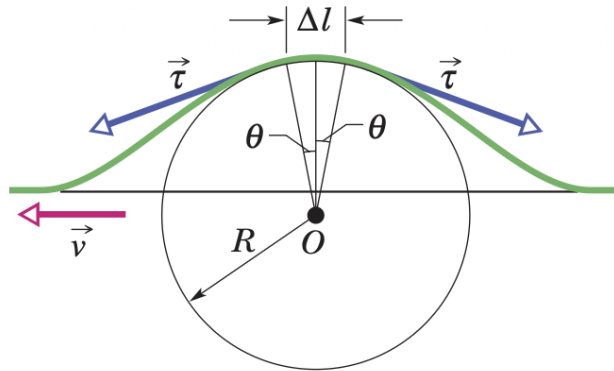
6 The Wave Speed, in a Different Way [Basics]

In class, we derived the wave equation from the microscopic dynamics $F = ma$ of a set of N -coupled oscillators, took the continuum limit $N \rightarrow \infty$ and got,

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu} \frac{\partial^2 y}{\partial x^2}, \quad (2)$$

as the equation describing transverse waves $y(x, t)$ on a taut string traveling around the x -direction. We identified $v = \sqrt{T/\mu}$ as the speed of the wave. Let us derive the wave speed in a somewhat different manner now.

To do so, consider a single symmetrical pulse such as that in the figure below, moving from left to right along a string with speed v . The tension in the string is τ . **For convenience, we choose a reference frame in which the pulse remains stationary**; that is, we run along with the pulse, keeping it constantly in view. In this frame, the string appears to move past us, from right to left in the figure with speed v .



- a Consider a small string element of length Δl within the pulse, an element that forms an arc of a circle of radius R and subtending an angle 2θ at the center of that circle. Find out the horizontal and vertical force components on this length element. Do one of these components add up to zero?
- b Now work with the small angle approximation to show that the **vertical force** on the string element is approximately $F \approx \tau \Delta l / R$.
- c At the moment shown in the figure, the string element Δl is moving in an arc of a circle. Apply $F = ma$ in the context of a centripetal force to show that the wave speed is indeed $v = \sqrt{T/\mu}$.